

COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 6

9 Logic and Proof (lp15)

- basic terminology (a) In the context of automatic theorem proving, provide one-sentence definitions of each of the following concepts: *satisfiable*, *sound*, *complete*. You may take as given the definitions of all underlying concepts. [3 marks]

Answer:

- A set of formulas or clauses is *satisfiable* provided an interpretation exists that makes all of them true.
 - A proof calculus or procedure is *sound* provided the formulas it designates as theorems are actually valid.
 - A proof calculus or procedure is *complete* provided it designates all valid formulas as theorems.
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- first-order logic, resolution theorem proving (b) Mordred has written a resolution theorem prover, but there are bugs in his code. Very rarely, one of the following errors occurs: a literal is deleted from a clause; an entire clause is deleted; the “occurs check” of unification is not performed. Briefly describe, with justification, the consequences of each type of error. [3 marks]

Answer:

- Deleting a literal destroys soundness: the prover could reach the empty clause when in reality the deleted literal is still present. Non-theorems could be “proved”. This error does not harm completeness however: consider that any resolution step on a particular literal effectively replaces that literal by zero or more literals from another clause.
 - Deleting a clause destroys completeness, since the deleted clause may be the only route to a proof. This error does not harm soundness however.
 - Not performing the occurs check can allow non-unifiable terms to be unified, resulting in unsound resolution steps, e.g. regarding quantifier reasoning. It could also lead to nontermination.
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- first-order logic, resolution theorem proving (c) For each of the following sets of clauses, either derive the empty clause or demonstrate that the set is satisfiable by exhibiting a model. Below, a and b are constants, while x , y and z are variables.

(i)

$$\begin{array}{ll} \{R(a)\} & \{\neg R(x), \neg Q(f(x)), \neg R(a)\} \\ \{Q(z), P(z)\} & \{\neg P(y), \neg R(y)\} \end{array}$$

[7 marks]

Answer: Using resolution, we easily derive $\neg Q(f(a))$, $P(f(a))$, $\neg R(f(a))$, and then reach a dead end. This suggests a model with integers for the universe and where

- a denotes 0
- $f(x)$ denotes $x + 1$

- $Q(x)$ and $R(x)$ denote $x = 0$
 - $P(x)$ denotes $x \neq 0$.
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(ii)

$$\begin{array}{ll} \{R(a), R(b)\} & \{\neg R(x), \neg Q(f(x)), \neg R(y)\} \\ \{Q(x), \neg P(y)\} & \{Q(z), P(z)\} \end{array}$$

[7 marks]

Answer: This proof will require the factoring rule in some form or other, because each clause has two or more literals. Simplest is first to resolve $\{Q(z), P(z)\}$ and $\{Q(x), \neg P(y)\}$ on the symbol P , yielding $\{Q(z), Q(x)\}$, which factors to $\{Q(z)\}$. Resolving this with $\{\neg R(x), \neg Q(f(x)), \neg R(y)\}$ yields $\{\neg R(x), \neg R(y)\}$, which factors to $\{\neg R(x)\}$. Two further resolution steps between this result and $\{R(a), R(b)\}$ yield the empty clause.

Note that resolving on two literals at once (as in the first two clauses) is completely wrong. For example, the set of clauses $\{R(a), R(b)\}$ and $\{\neg R(a), \neg R(b)\}$ is satisfiable.
